

NNTApp:

A MATLAB application for the flux calculation of the NNT-mediated reaction

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Overview

NNTApp is used to calculate the flux mediated by nicotinamide nucleotide transhydrogenase (NNT), an enzyme located at mitochondrial inner membrane which transfers a hydride between NAD(H) and NADP(H).

The enzymatic kinetics of NNT is derived from its molecular mechanism. Following enzyme kinetics, a mathematical model is developed. We implemented the model using MATLAB. To facilitate the usage of this model, we developed this graphical user interface application.

The application is developed with MATLAB 2022a.

Installation

Windows

The applications for windows were compiled using MATLAB R2022a.

Two installation methods, installer and redistribution, are supplied to install this application.

1) Installer

step 1.

Download the installer

step 2.

Start the installation by double clicking installer.exe.

Once the installation is completed, you are able to run NNTApp.

The installer contains MATLAB runtime for 2022a. No additional installation or downloading is required.

2) Redistribution

step 1.

Download the redistribution folder

step 2.

Install MATLAB runtime for MATLAB 2022a in your computer following the instructions in the file readme.txt.

step 3.

Double click NNTApp.exe in the redistribution folder to run the application.

Ubuntu

The applications for ubuntu were compiled in MATLAB R2023b.

Two installation methods, installer and redistribution, are supplied to install this application.

1) Installer

step 1.

Download the installer folder

step 2.

Start the installation by double clicking installer.exe.

- i. assign the installation directory for NNTApp.exe
- ii. assign the installation directory for MATLAB runtime, and record the path
(you will need the path when you run the applicaiton)
- iii. complete the installation

step 3.

- i. open terminal
- ii. move to the directory where NNTAPP.exe is installed.
- iii. run the script by the command
`./run_NNTApp.sh your_runtime_path`
your_runtime_path is the directory that you assigned when you install the application.
e.g. `/usr/local/MATLAB/R2023b`

2) Redistribution

step 1.

Download the redistribution folder for ubuntu

step 2.

Install MATLAB runtime for MATLAB 2023b in your computer following the instructions in the file readme.txt.

step 3.

i. open terminal

ii. move to the directory where NNTAPP.exe is installed.

iii. run the script by the command

```
./run_NNTApp.sh your_runtime_path
```

your_runtime_path is the directory where MATLAB runtime located. e.g.

```
/usr/local/MATLAB/R2023b
```

Calculate: NNT flux

We can estimate the flux of the NNT-mediated reaction with the assigned parameters using the kinetic model for NNT.

The parameters are listed in the left panel. You can adjust the parameters and then click ‘Go!’ button in the ‘NNT flux’ tab page. The application will calculate the flux and show the result on the ‘NNT flux’ tab page.

The definition of parameters is listed in the below table.

Parameter	Definition	Unit	Default
NADH	The concentration of NADH in the medium where NNT works.	μM	200
NAD ⁺	The concentration of NAD ⁺ in the medium where NNT works.	μM	400
NADPH	The concentration of NADPH in the medium where NNT works.	μM	38

NADP ⁺	The concentration of NADP ⁺ in the medium where NNT works.	μM	2
ΔE ₀	The standard redox potential difference of NAD(H) and NADP(H)	mV	0
ΔΨ _m	Mitochondrial membrane potential	mV	-150
pH _o	The pH value in the medium where NNT uptakes H ⁺ for the forward reaction, e.g. inter membrane space	-	6.88
pH _i	The pH value in the medium where NNT transports H ⁺ to for the forward reaction, e.g. mitochondrial matrix	-	7.78
Temp	The temperature in the medium where NNT works	K	310.15
V _{max}	The maximum capacity of NNT. The value depends on the amount and status of NNT protein	μmol/min	1
K _{NADH}	The Michaelis-menten constant of NADH for the NNT-mediated reaction	μM	10
K _{NAD⁺}	The Michaelis-menten constant of NAD ⁺ for the NNT-mediated reaction	μM	125
K _{NADPH}	The Michaelis-menten constant of NADPH for the NNT-mediated reaction	μM	25
K _{NADP⁺}	The Michaelis-menten constant of NADP ⁺ for the NNT-mediated reaction	μM	17
V _i	The volume of the medium where NNT works, e.g. mitochondrial matrix	mL	10

Note: Since parameters are fixed, V_i in this case doesn't play a role.

Calculate: temporal process

We can estimate how the changes on the substrates with assigned initial substrate concentrations and other parameters. As to this application, we assume:

- the amounts of substrates (NADH + NAD⁺, and NADPH + NADP⁺) are constant;
- other parameters, e.g. ΔΨ_m, pH_i, pH_o, are stable.

iii. NNT is the only enzyme working for the metabolism of NAD(H) and NADP(H).

In most biological systems, e.g. an intact mitochondrion or an intact cell, assumption iii is not true. So, this calculation mainly helps the understanding of how NNT can contribute to NAD(H) and NADP(H) metabolism rather than a simulation of the metabolism of NAD(H) and NADP(H) in a biological system.

The definition of parameters is the same to that for flux calculation. You can adjust the parameters in the parameter panel and then click 'Go!' button in the 'Temporal process' tab page. The application will calculate the temporal changes of the NNT-mediated flux, NADH concentration, NAD^+ concentration, NADPH concentration and NADP^+ concentration. The results are shown on the 'temporal process' tab page as figures.

To calculate the changes on substrate concentrations, V_i is required. In a lot of cases, $V_{\max}$ is relevant to V_i . A bigger volume usually contains more NNT enzyme proteins and correspondingly a bigger $V_{\max}$. It is not mandatory, but it is good to check the other when you modify one of $V_{\max}$ or V_i .